

IPsec VPN Executive Assessment

Audited Target: ikev2_s2s_ipsec_vpn_aes_gcm.pcapng | Captured Packets: 12

Audit Date: 29 Sep 2026, 23:32 UTC • Regulatory Baseline: NIST SP 800-77 / RFC 7296

SECURITY RISK SCORE 5/100 LOW Rating	AI TRAFFIC CLASS P2P XGBoost Inference	CLASSIFIER CONFIDENCE 41.3% Softmax Probability	CIPHER SUITE AES-256-GCM AEAD
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3x3 Threat Assessment Matrix

	LOW IMPACT	MED IMPACT	HIGH IMPACT
HIGH LIKELIHOOD	MEDIUM	HIGH	CRITICAL
MED LIKELIHOOD	LOW	MEDIUM	HIGH
LOW LIKELIHOOD	LOW	LOW	MEDIUM

Cryptographic & Protocol Compliance Parameters

PARAMETER	OBSERVED VALUE IN CAPTURE	COMPLIANCE BASELINE (NIST)
IPsec Protocol / IKE Version	IKEv2 (IKE_AUTH)	IKEv2 (RFC 7296)
Encryption Cipher & Key Size	AES-256-GCM (256-bit)	AES-256-GCM / AES-256-CBC
Integrity & Hash Algorithm	AEAD	AEAD / HMAC-SHA2-256
Diffie-Hellman Key Exchange	Group 19	Group 14, 19, 20, 21, 28 (≥ 2048-bit)
Security Association (SPIs)	Init: 62f94c9887715a34 Resp: f05534eef373dc1b	Valid 64-bit Non-Zero SPI Pair

Security Assessment & Actionable Recommendations

[IPSEC-META-003] Static SPI Header Linkability / Session Tracking Risk [LOW] Observed: Observable ESP SPI(s): 0x5fbd60dc, 0xa70bbb39 Expected: Dynamic SA Rekeying & SPI Rotation Remediation: Ensure frequent Security Association rekeying to prevent passive netflow session correlation.
